

MATTHEW HERNANDEZ

US Citizen | 954-736-6926 | matthewh0729@gmail.com | LinkedIn: in/mh0729/ | Eportfolio: matthewh007.github.io

EDUCATION

University of South Florida (Tampa, FL)
Bachelor of Science in Aerospace Engineering

Graduating: May 2027
GPA: 3.69

EXPERIENCE

Jet Propulsion Laboratory (La Cañada Flintridge, CA)
NASA Europa Clipper ICONS Intern

June 2025 - August 2025

- Retrofitted and assembled a vacuum system with additional spectral accessories to replicate Europa-like environmental conditions, enabling the irradiation and analysis of sulfate compounds in salt-brine samples.
- Implemented **Python scripts** to process data from CASINO Monte Carlo simulations, analyzing electron stopping energies across materials to provide theoretical benchmarks for experimental comparisons.
- Interpreted **IR and UV-Vis spectroscopy** data with **MATLAB** scripts to generate datasets for the Europa Clipper mission.

KULR Technology Group (Houston, TX)
Engineering Intern - Prototyping and Testing

February 2025 - May 2025

- Applied **Python** scripts to integrate **computer vision** for an image analysis study, improving data retention by **40%**.
- Optimized **20+** CAD drawings in SolidWorks to meet industry **GD&T** standards, ensuring accurate documentation and efficient data management within SolidWorks PDM.
- Performed cycle life testing using programmable power supplies, executing **100+** charge/discharge cycles while monitoring voltage, current, and temperature for flight reliability verification.

NASA KSC - Applied Chemistry Lab (Kennedy Space Center, FL)
OSTEM Engineering Intern

August 2024 - December 2024

- Contributed to troubleshooting issues of a **pure oxygen gas system** by fixing leaks and restructuring the flow system on **LabVIEW**, reducing downtime by **20%** and increasing overall project productivity.
- Supported experimentation with **vacuum chambers** for projects like **Molten Regolith Electrolysis (MRE)** and outgassing characterization of heating lunar simulants for applications of lunar base habitability.
- Led vacuum environment setup and off-gassing analysis of lunar simulants using a **Residual Gas Analyzer (RGA)** for research for different clients to assess gas composition.

Cornell Astrophysics and Planetary Science REU (Ithaca, NY)
Research and Development Intern

June 2024 - August 2024

- Designed and fabricated a water-cooling loop with PVC and Silicon Tubing for a prototype **Hall Effect Thruster**, enabling regenerative cooling experiments for power cycles ranging from kW to MW.
- Performed preliminary **ANSYS simulations** to estimate thermal behavior and inform design decisions for regenerative cooling.
- Utilized **SolidWorks** to create 6 test stand iterations for the thruster, producing a **15%** improvement in structural and thermal stability compared to previous models.

LEADERSHIP EXPERIENCE

Society of Hispanic and Professional Engineers USF (Tampa, FL)
Professional Development Workshop and Resume Director

June 2025 - Present

- Planning and initiating professional development workshops for **400+** members on topics such as resumes, mock interviews, and conference readiness for the upcoming SWE and SHPE national conferences.
- Managed cross-club event planning by organizing communication channels, and tracking logistics to strengthen SHPE USF programming and expand chapter reach across campus.

Institute of Electrical and Electronics Engineers UCF (IEEE) (Orlando, FL)
Professional Development Chair

June 2024 - December 2024

- Organized events for **250+** members on topics such as resumes, mock interviews, and conference readiness, increasing turnout by **30%** compared to the previous year.
- Delegated a committee of **5** students to structure and gather resources for a mentorship program and ProDev events.

SKILLS SUMMARY

Software: SolidWorks, LabVIEW, Siemens NX, Ansys, CASINO

Technical Skills: Computer Aided Design (CSWA Certified), R&D Experimentation, Computer Vision Models, GD&T Optimization, Finite Element Analysis (FEA), 3D Printing (AMA Certified), Soldering, Spectrometry Analysis, Simulation Design

Languages: Python (Advanced), MATLAB (Intermediate), C (Intermediate), CSS (Intermediate), HTML (Intermediate)